

COMPUTATIONAL INVESTIGATION OF WATER **ADSORPTION IN SILICA**

SUBMITTED TO

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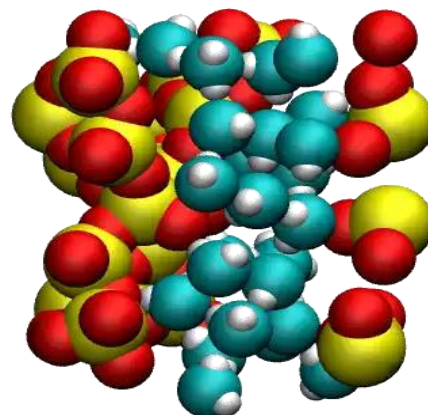
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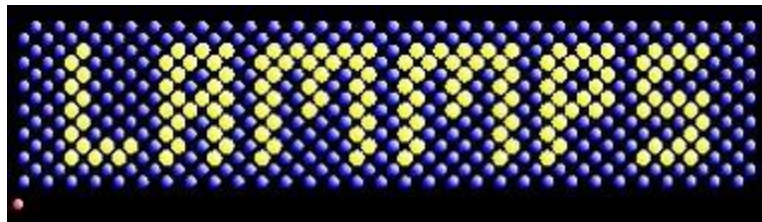
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Introduction This report explains the simulation of water adsorption in silica using LAMMPS software. Adsorption is the process where molecules from a gas or liquid stick to a surface. Silica (SiO_2) is widely used in adsorption-related studies because of its large surface area and chemical properties. Understanding how water behaves on silica surfaces is important in many fields like environmental science, catalysis, and material development. This tutorial helped us to simulate such a process and analyze the molecular-level interactions using Molecular Dynamics (MD) simulations.



Objective of the Simulation

The main objective of this simulation was to model and study how water molecules interact with a silica surface.

By doing so, we could visualize and analyze the adsorption process, including how water molecules move, bond, or cluster near the silica surface.

Specifically, we aimed to:

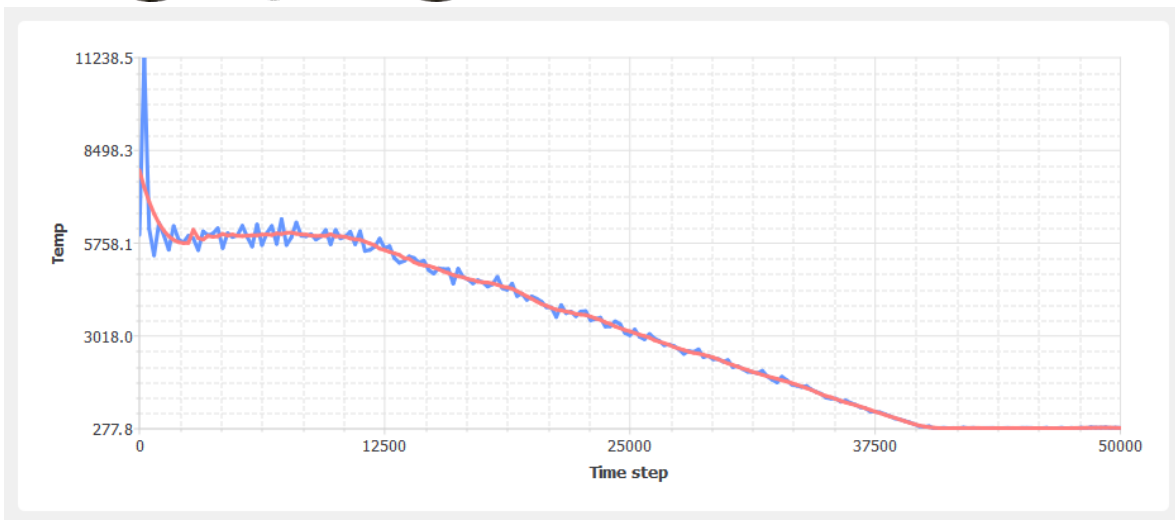
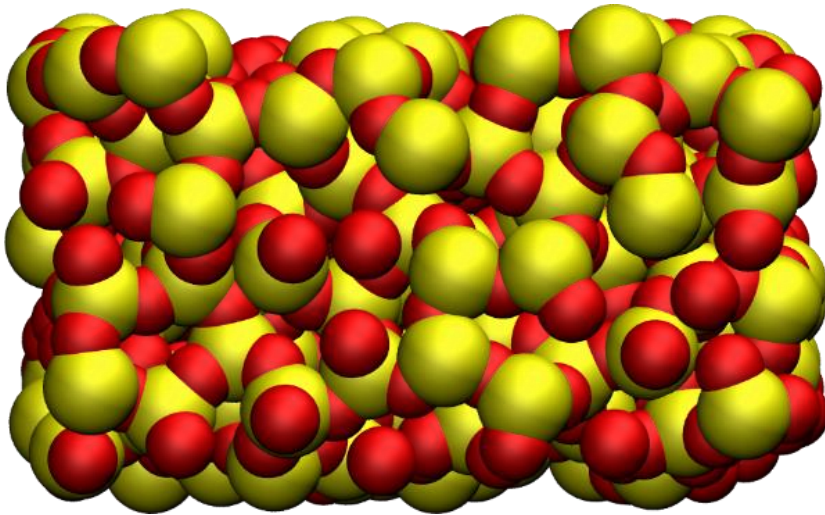
- Learn how to use LAMMPS for surface and molecule interactions
- Understand the setup required for water adsorption studies
- Observe and analyze the adsorption behavior of water on silica
- Use appropriate force fields to represent reactive interactions

Preparing the Silica Structure The simulation began by preparing the silica structure. Instead of generating it from scratch, a pre-made data file named silica.data was used. This file contained atom positions and bonding information for silicon (Si) and oxygen (O) atoms. The silica model used was amorphous, meaning it had a disordered structure like glass, rather than a regular crystal structure. This more accurately represents real-life silica surfaces used in adsorption studies.

The structure was visualized using OVITO, where we could confirm the presence of a solid silica layer and identify the space above it where water molecules would later be inserted. create a new folder and add a file named **generate.lmp**.

They providing input script in **LAMMPS tutorial**. Used to run the script file in **LAMMPS GUI**.

Figure: Amorphous silica (SiO_2). Silicon atoms are represented in yellow, and oxygen atoms in red.

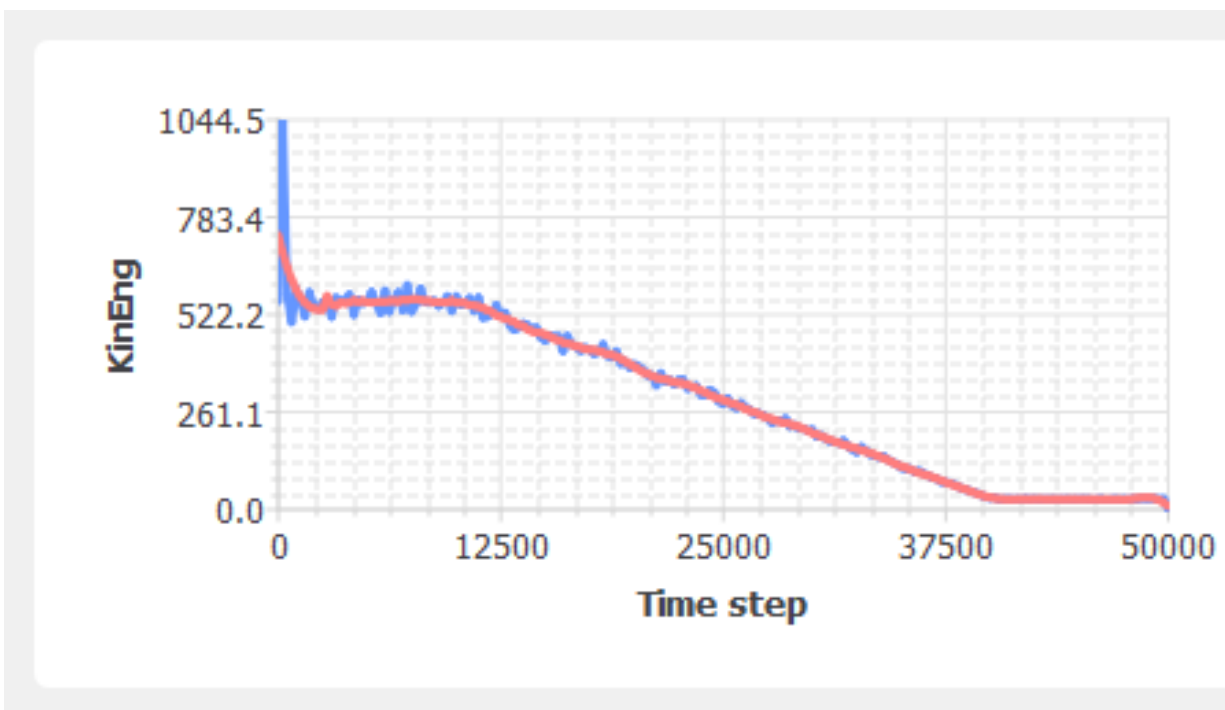


a) Temperature, T , as a function of time, t , during the annealing of the silica system. b) System density, ρ , during the annealing process. The vertical dashed lines mark the transition between the different phases of the simulation.

Density vs Time



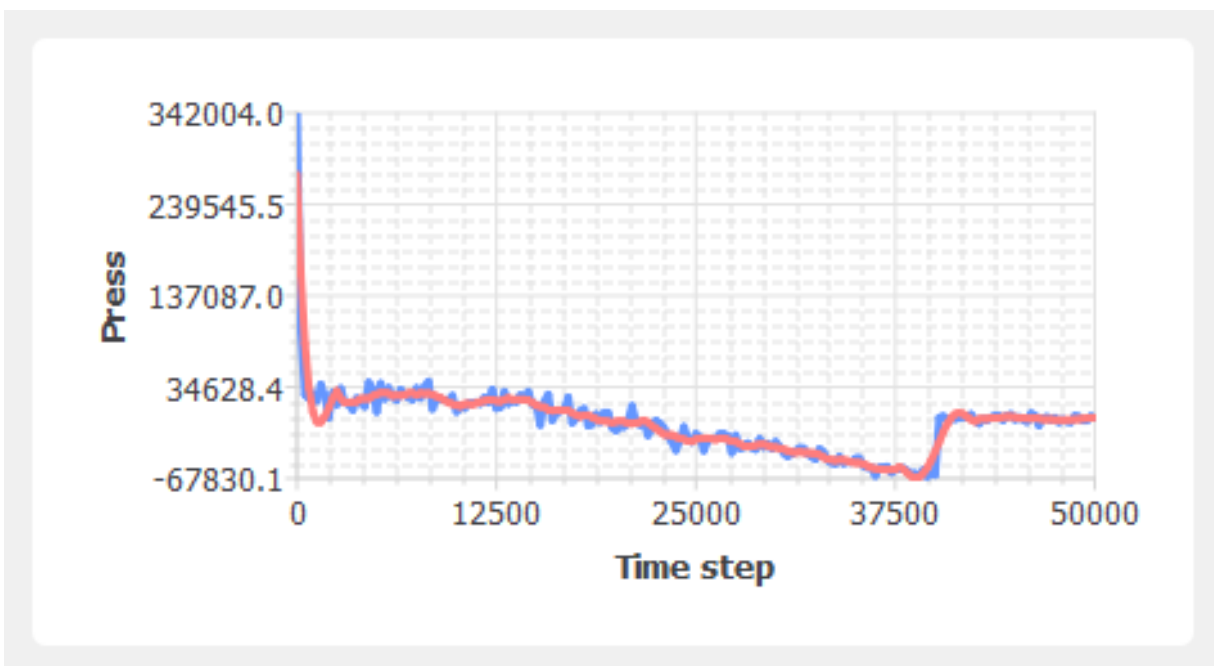
Kinetic energy vs Time



Potential energy vs Time



Pressure vs Time



Total energy vs Time



Run the simulation using LAMMPS. From the Charts window, the temperature evolution can be observed, showing that it closely follows the desired annealing procedure. The evolution of the box dimensions over time confirms that the box deformed during the last stage of the simulation. After the simulation completes, the final LAMMPS topology file called **generate.data** will be located next to **generate.lmp**.

Defining the Simulation Box and Regions The simulation box dimensions were adjusted to create enough space above the silica slab. This empty region allows water molecules to be added without initial overlaps. Periodic boundary conditions (PBC) were applied in the x and y directions, meaning atoms leaving one side of the box re-enter from the opposite side. In the z-direction, PBC was turned off to mimic a surface exposed to vacuum or air.

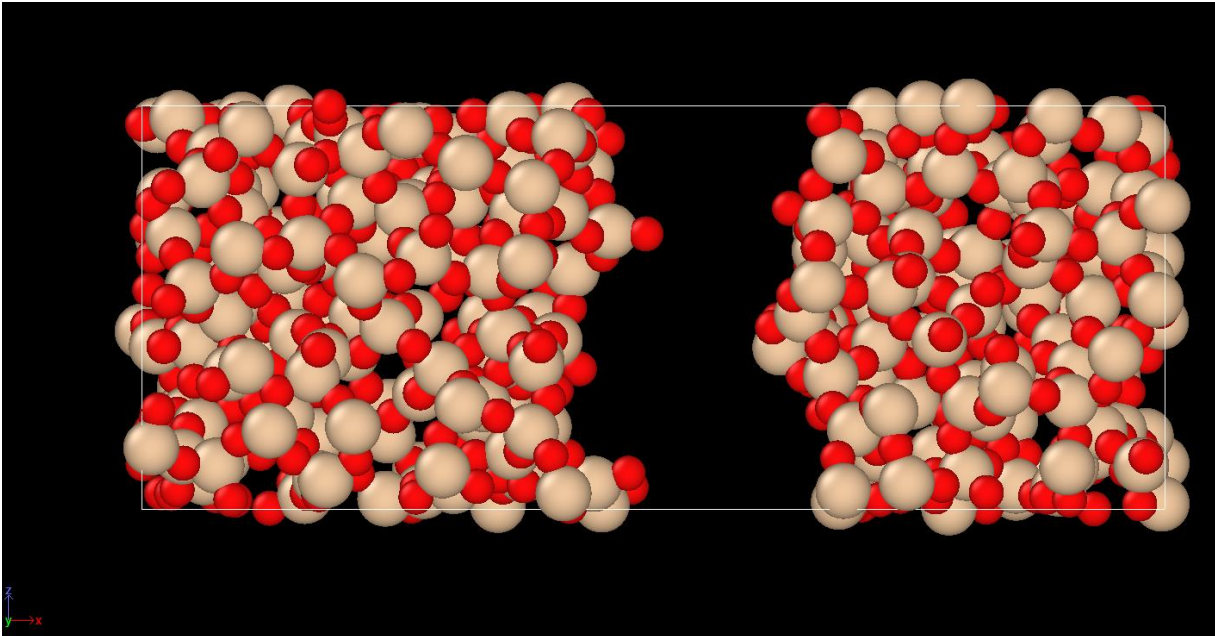
We also defined specific regions using the `region` command in LAMMPS. This helps to insert molecules in a controlled area and avoid overlaps with the existing silica atoms.

CRACKING THE SILICA

Create a new file called **cracking.lmp**,

That one command **read_data generate.data**

The **generate.data** is copy and paste in cracking folder. And run it script.



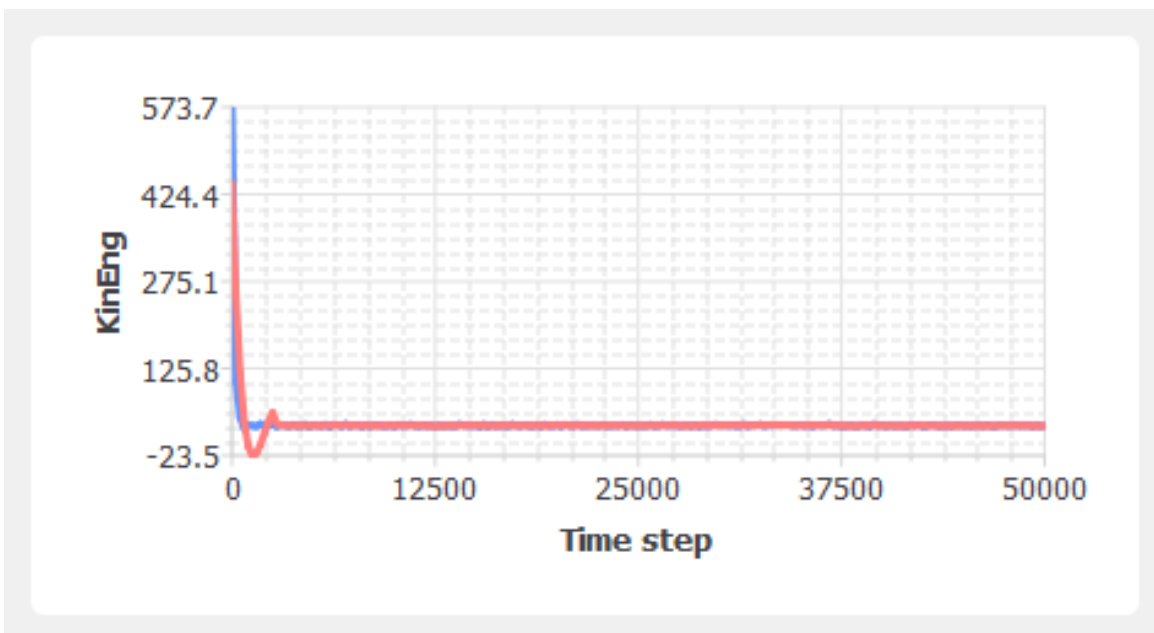
Temperature vs Time



Density vs Time



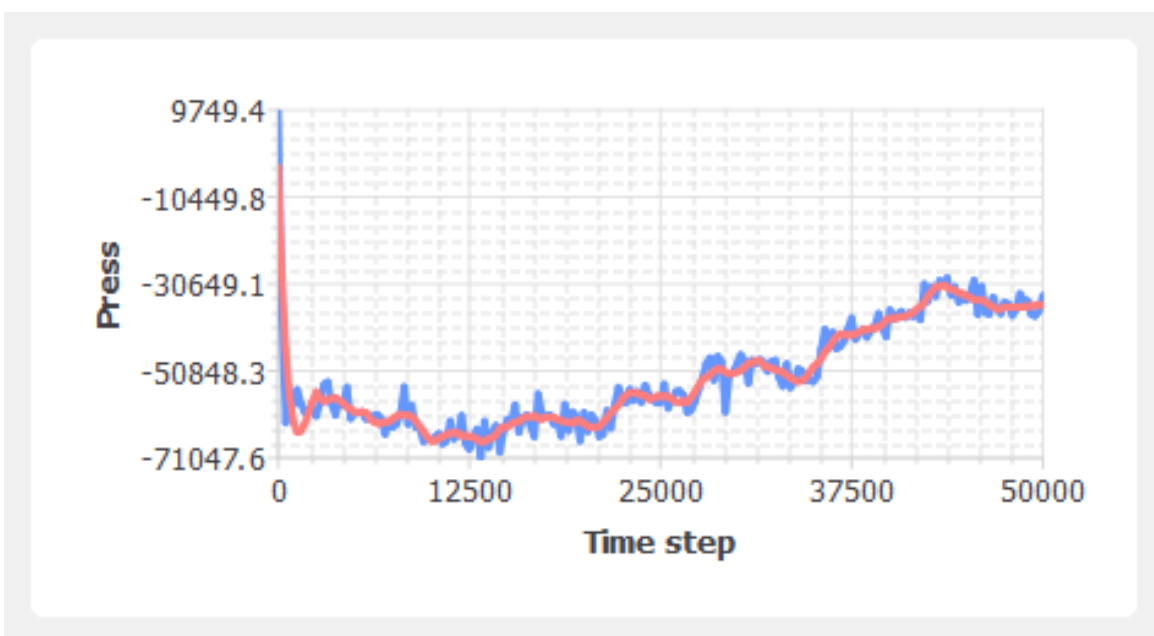
Kinetic energy vs Time



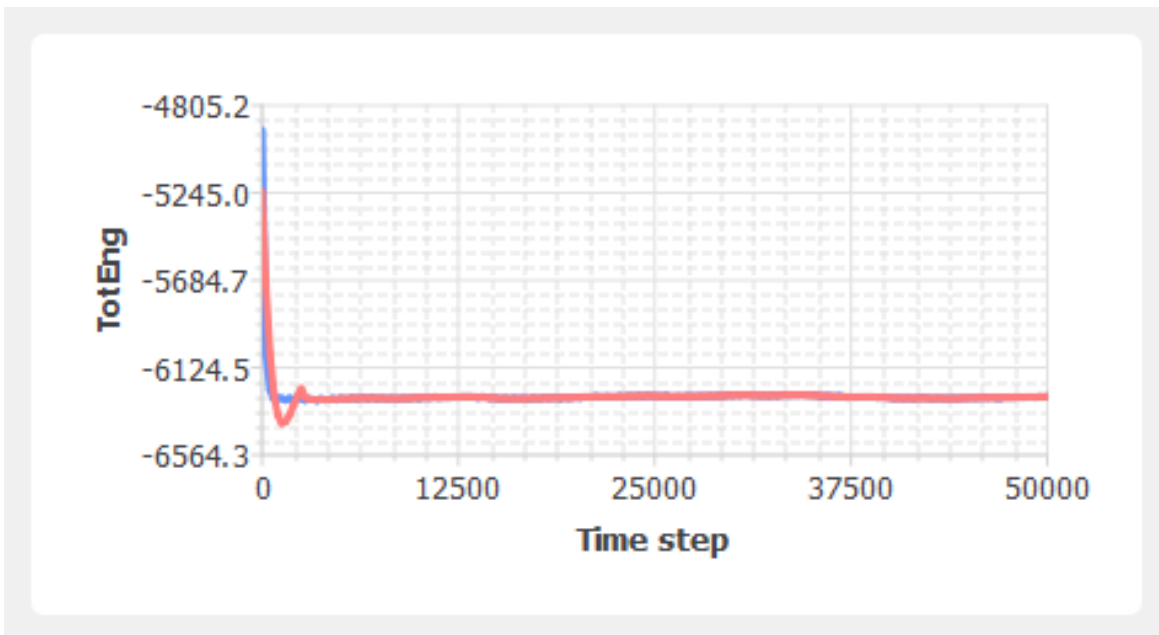
Potential energy vs Time



Pressure vs Time



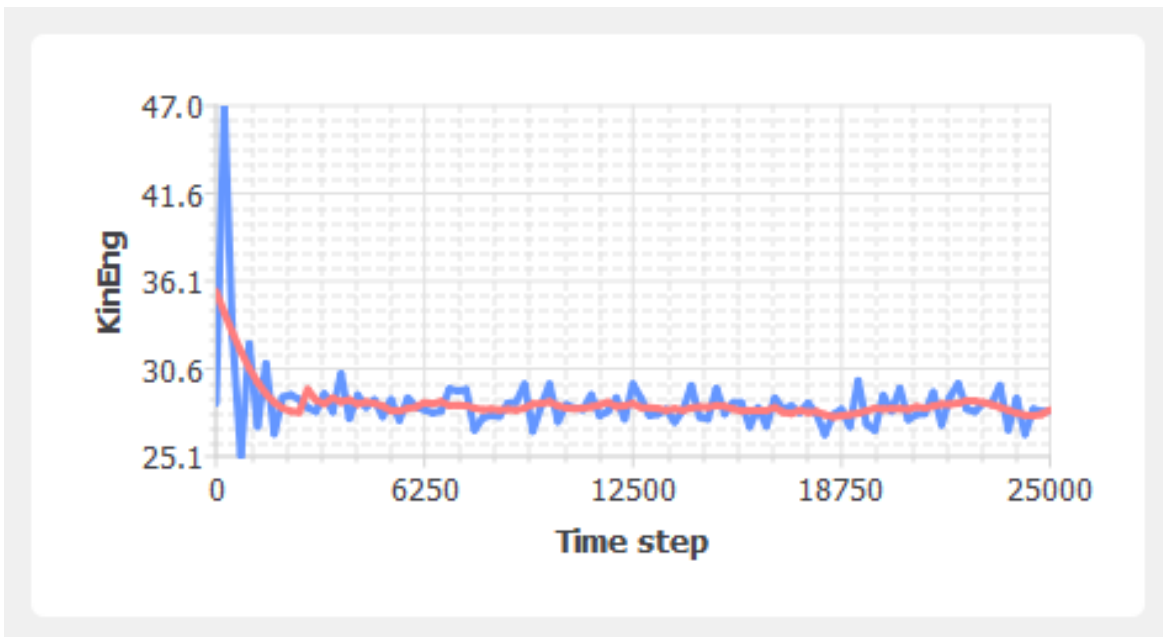
Total energy vs Time



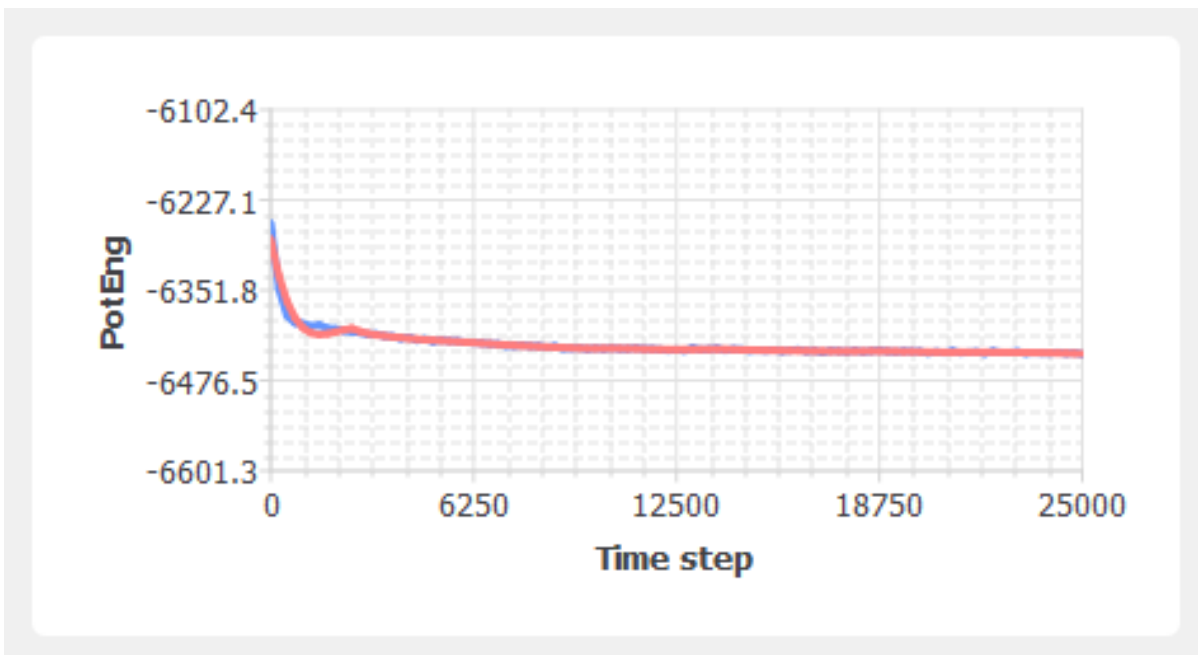
Adding Water Molecules



Kinetic energy vs Time



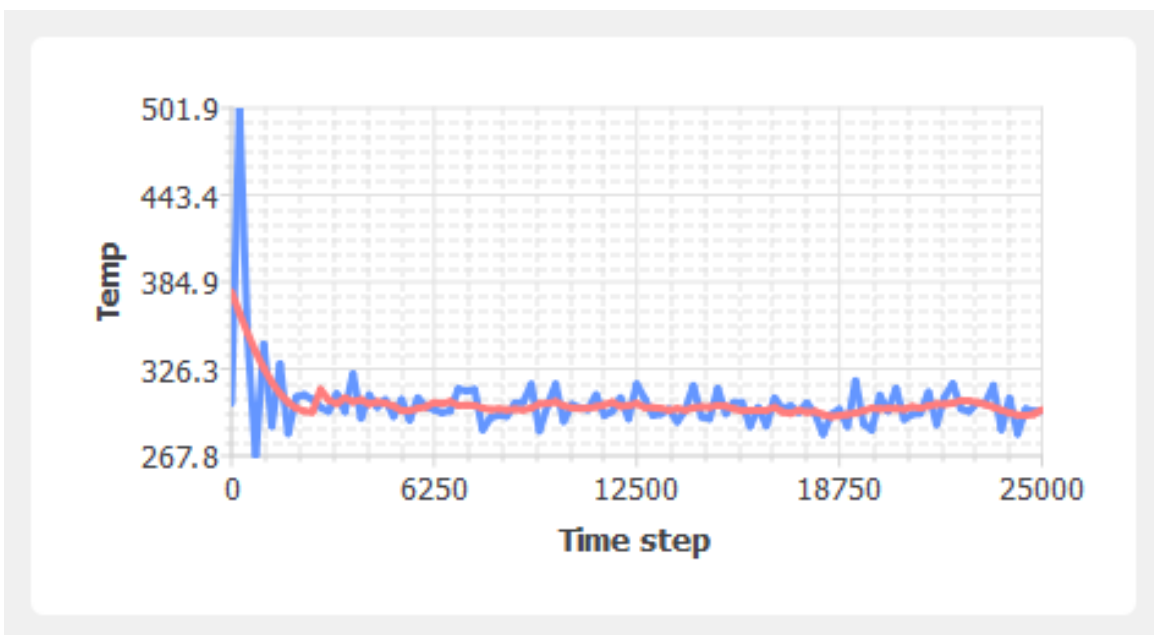
Potential energy vs Time



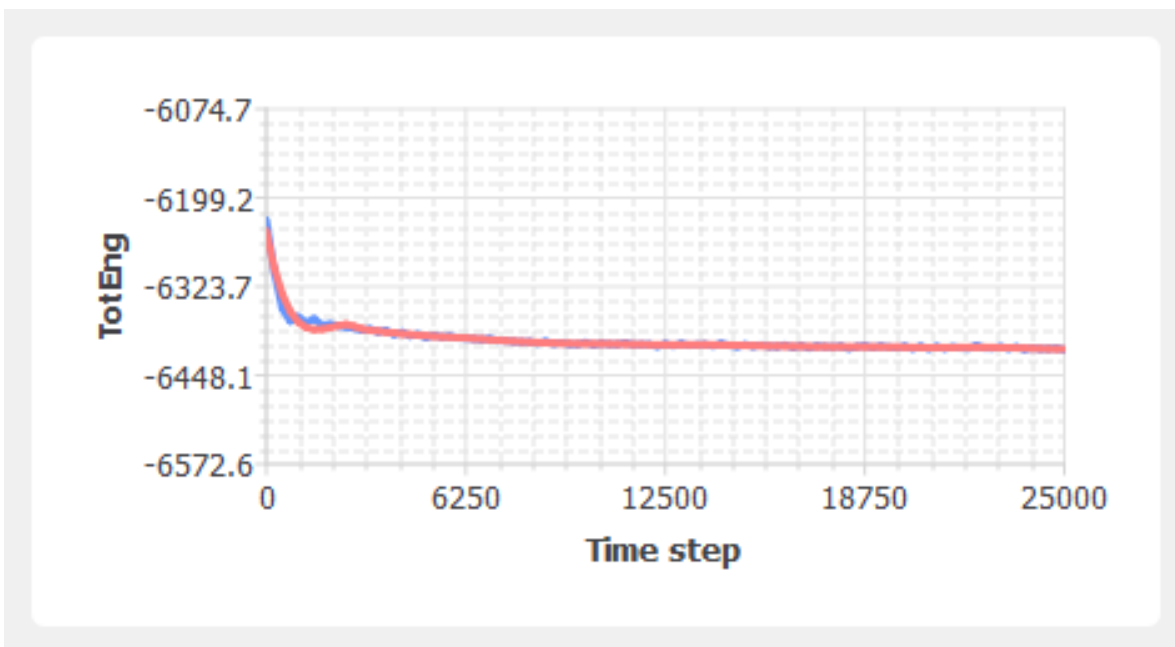
Pressure vs Time



Temperature vs Time

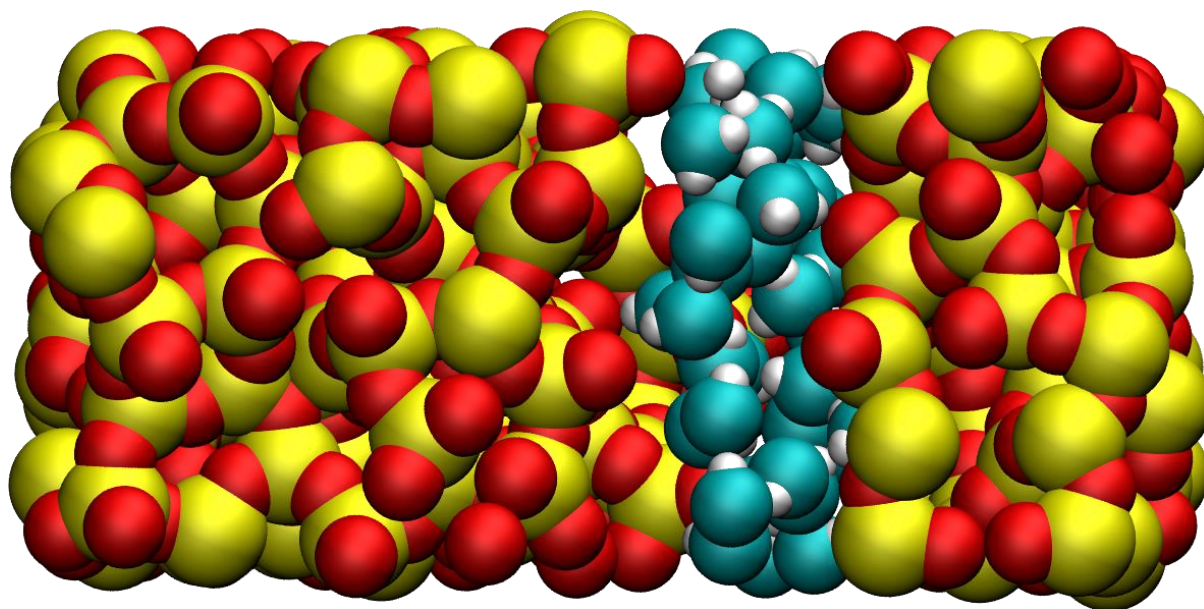


Total energy vs Time



Water molecules were added into the simulation using a molecule template called H2O.mol , which defines the geometry and atomic types of water (1 oxygen, 2 hydrogen atoms). The command `molecule h2omol H2O.mol` was used to load the water structure, and then `create_atoms` placed multiple water molecules randomly in the vacuum region above the silica surface.

We ensured that there were no overlaps using overlap options and randomized placement to simulate a realistic distribution. This step was important to mimic how water molecules would behave when approaching the silica surface.



Running the Simulation (Equilibration) The system was first equilibrated using the NVT ensemble (Number, Volume, Temperature constant). This ensures that the temperature is stabilized at 300 K and the system reaches a realistic starting point. The timestep used was 0.25 femtoseconds, and the simulation was run for several thousand steps (e.g., 10,000).

The commands used were:

```
fix mynvt all nvt temp 300.0 300.0 100.0  
run 10000
```

This gradually allowed the system to reach thermal equilibrium, and we could then monitor how water molecules behaved in the presence of the silica surface.

Observations and Visualization After running the simulation, we used visualization tools like OVITO to view the results. The water molecules were seen moving closer to the silica surface. Some molecules formed hydrogen bonds with surface oxygen atoms, while others stayed in the bulk phase.

This part of the tutorial helped us see how MD simulations can be used to study molecular interactions and adsorption behavior. We could also analyze changes in atom positions, velocities, and bonding throughout the simulation time.

Learning Outcomes Through this tutorial, we learned: - How to use LAMMPS to simulate water adsorption on silica - How to import and use molecule templates in a simulation - How to define and visualize surface systems using simulation tools - How to analyze molecular movement and adsorption processes

This knowledge can be applied in many areas such as surface chemistry, catalysis, water purification, and environmental engineering, fluid mechanics.

Conclusion In conclusion, this LAMMPS tutorial successfully demonstrated the water adsorption process on an amorphous silica surface. The simulation setup, execution, and analysis were simple yet powerful tools to understand surface-molecule interactions at the atomic level. The tutorial improved our understanding of using Molecular Dynamics in research and industrial applications. We are now more confident in performing similar simulations for other materials and adsorption studies.